Design and Development of Drinking Water Quality Alert System with Voice Announcement for Remote Areas

A Project Report submitted in partial fulfillment of the requirements for the Award of Degree of

Bachelor of Technology

In

Computer Science and Engineering (Internet of Things)

By

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B.Bharath Kumar	2211CS050015
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Under the esteemed guidance of

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School of Engineering

MALLA REDDY UNIVERSITY

Maisammaguda, Dulapally, Hyderabad, Telangana 500100

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(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14, Higher Education (UE) Department)

Department of Computer Science & Engineering (Internet of Things)

CERTIFICATE

This is to certify that the project report entitled **"Design and Development of Drinking Water Quality alert System With Voice Announcement for Remote Areas",** submitted by **A.Haresh (2211CS050001)**, **B.Bharath (2211CS050015), G.Ganga reddy (2211CS050028),** towards the partial fulfillment for the award of Bachelor's Degree in Computer Science and Engineering – Internet of Things, Malla Reddy University, Hyderabad, is a record of bonafide work done by him/ her. The results embodied in the work are not submitted to any other University or Institute for award of any degree or diploma.

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DECLARATION

We hereby declare that the project report entitled **"Design and Development of Drinking Water Quality Alert System with Voice Announcement for Remote Areas"** has been carried out by us. This work has been submitted to the **Department of Computer Science and Engineering (Internet of Things), Malla Reddy University, Hyderabad** in partial fulfillment of the requirements for the award of degree of Bachelor of Technology. We further declare that this project work has not been submitted in full or part for the award of any other degree in any other educational institutions.

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ABSTRACT

Ensuring safe drinking water is a critical public health requirement, especially in the present scenario where water pollution and contamination are increasing rapidly due to industrialization, urbanization, and improper waste disposal. Traditional methods of water quality testing are mostly manual, time-consuming, and require laboratory analysis, which makes them inefficient for continuous monitoring. Moreover, these conventional approaches do not provide real-time data, making it difficult to take immediate action in case of contamination. To address these limitations, this project focuses on the development of an advanced Drinking Water Quality Monitoring and Alert System that enables continuous, real-time monitoring of water quality. The system integrates multiple sensors to measure key water quality parameters such as pH level, turbidity, total dissolved solids (TDS), and temperature. These parameters play a vital role in determining whether water is safe for drinking and domestic use.The sensors are connected to a Wi-Fi enabled microcontroller, such as an ESP32, which acts as the core processing unit of the system. It collects data from the sensors at regular intervals and transmits it to a cloud- based platform using Internet of Things (IoT) technology. The cloud platform stores the data and provides visualization through dashboards, allowing users to monitor water quality remotely using a smartphone or computer.The system continuously compares the measured values with standard safe drinking water limits defined by health organizations. If any parameter exceeds the permissible range, the system immediately generates alerts and notifications, which are sent to users or concerned authorities through mobile applications or SMS. This early warning mechanism helps in taking quick preventive measures, thereby reducing the risk of water-borne diseases.

The proposed system is cost-effective, easy to deploy, and scalable, making it suitable for both urban smart city applications and rural areas where access to safe drinking water is limited. It minimizes human intervention, enhances accuracy, and ensures reliable monitoring of water quality.

Keywords: IoT, Water Quality Monitoring, ESP32, pH Sensor, Turbidity Sensor, TDS Sensor, Temperature Sensor, Real-Time Monitoring, Cloud Computing, Blynk, Voice Alert System, Smart Water Management

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CHAPTER-1

INTRODUCTION

1.1 Problem Definition & Description

Access to safe drinking water is a major problem in many rural and remote areas. People often use water from wells, rivers, or tanks without knowing whether it is safe to drink. Contaminated water can cause serious health problems and diseases. Traditional water testing methods require laboratories and trained professionals, so they are not suitable for regular monitoring. The proposed system monitors water quality using sensors and a microcontroller. It also gives a voice announcement alert to inform users when the water is unsafe for drinking.

1.2 Objectives of the Project

The main objective of this project is to design and develop a system that can monitor the quality of drinking water in real time. The system measures important water parameters such as pH, turbidity, and temperature using sensors. It detects unsafe water conditions and automatically generates alerts when the water quality crosses safe limits. The system also provides voice announcements to inform users whether the water is safe or unsafe for drinking. This project aims to create a low-cost, easy-to- use solution suitable for rural and remote areas, helping to improve public health by ensuring access to safe drinking water.

1.3 Scope of the Project

The scope of this project is to design and develop a smart system that continuously monitors the quality of drinking water using sensors and a microcontroller. The system measures important parameters such as pH, turbidity, and temperature to determine whether the water is safe for consumption. When the water quality crosses the safe limits, the system provides an alert through a voice announcement so that users can easily understand the water condition.

This project is mainly intended for rural and remote areas where regular laboratory testing of water is difficult and expensive. The system helps people quickly identify unsafe drinking water and take necessary precautions. It can be installed in water storage tanks, wells, or community water supply systems to provide continuous monitoring.

CHAPTER-2

SYSTEM ANALYSIS

2.1 Existing System

2.1.1 Background & Literature Survey

Water is one of the most essential resources for human life, and maintaining its quality is very important for health and safety. In many rural and remote areas, people depend on wells, borewells, rivers, and storage tanks for drinking water. However, these sources may be contaminated due to physical, chemical, or biological impurities. Unsafe drinking water can lead to various diseases and health issues, making regular water quality monitoring necessary.

Traditional methods of water quality testing are usually performed in laboratories using specialized equipment. Although these methods provide accurate results, they are costly, time-consuming, and not suitable for continuous monitoring in remote locations. Because of these limitations, researchers have focused on developing smart and low-cost water quality monitoring systems using embedded systems and IoT technologies.

From the literature survey, it is observed that many existing systems use sensors to measure water parameters such as pH, turbidity, temperature, and TDS. These systems commonly use microcontrollers like Arduino, ESP32, or Raspberry Pi to collect and process sensor data. Some systems also send the data to cloud platforms or mobile applications for remote monitoring. Research studies show that IoT-based water quality monitoring systems are effective in providing real-time information and improving water management.

Recent advancements in embedded systems and Internet of Things (IoT) technologies have made it possible to develop smart water quality monitoring systems. Several research studies have proposed systems that use sensors such as pH sensors, turbidity sensors, temperature sensors, and Total Dissolved Solids (TDS) sensors to measure different water parameters. These sensors are connected to microcontrollers like Arduino, ESP32, or Raspberry Pi, which collect and process the data. The processed data can then be displayed on screens or transmitted to cloud platforms and mobile applications for remote monitoring.

Many existing systems focus on providing real-time monitoring and sending alerts through SMS, mobile applications, or web dashboards. These systems help authorities and users monitor water conditions efficiently. However, most of these solutions require smartphones, internet connectivity, or the ability to read digital displays. In rural areas, many users may not have access to smartphones or may not be able to understand text-based notifications.

To overcome these limitations, the proposed system introduces a voice announcement feature that directly informs users about the safety of drinking water. The system continuously monitors water parameters and compares them with standard safety levels. When the water quality exceeds safe limits, the system generates an audio warning through a speaker. This approach makes the system more user-friendly and accessible, especially for people living in remote areas**.**

Several researchers have proposed IoT-based solutions for monitoring water quality parameters such as pH, turbidity, and total dissolved solids (TDS). Konde and Deosarkar [1] developed an IoT-based monitoring system that enables real-time data collection and remote monitoring. Similarly, Sugapriyaa et al. [2] designed a smart system for real-time applications, emphasizing the importance of continuous monitoring for improving water safety. Unnikrishna Menon et al. [3] introduced a wireless sensor network-based approach for monitoring river water quality, demonstrating the effectiveness of distributed sensing systems.

Kumar and Samalla [4] proposed a system integrating IoT technologies for efficient monitoring and analysis of water parameters. Hamid et al. [5] further enhanced system performance by evaluating IoT-based water monitoring systems with improved accuracy and reliability. Kumar and Sinha [6] developed an automatic monitoring system using IoT, highlighting reduced human intervention and faster response times.

Recent studies have also focused on integrating cloud platforms and data analytics. Arumugam et al. [7] and Rahmani and Shishir [8] emphasized the role of IoT and cloud computing in enabling real-time data transmission and storage. Pande et al. [9] and Kumar et al. [10] demonstrated the use of IoT systems for efficient monitoring and decision-making through cloud-based platforms. Ramesh et al. [11] extended the application of IoT monitoring systems to aquaculture, showing versatility in different environments.

Further advancements include intelligent monitoring systems using advanced sensors and predictive techniques. Thangavelu and Stalin [13] proposed an intelligent system using IoT and nano sensors for enhanced accuracy. Kumar and Dhiman [14] introduced an integrated IoT-based system for smart water management. Sureshkumar and Muralidharan [15] developed a smart monitoring system focusing on real-time data acquisition and system efficiency.

2.1.2 Limitations of Existing System

Fig. 2.1.2 shows the water quality testing process based on manual sampling and laboratory analysis. The current approach involves collecting water samples from various sources and analyzing them in laboratories to determine key quality parameters such as pH, TDS, and turbidity. Although this method provides accurate results, it is labor-intensive and time-consuming. Moreover, it does not support real-time monitoring, and there is no alert system to notify users in case of contamination.

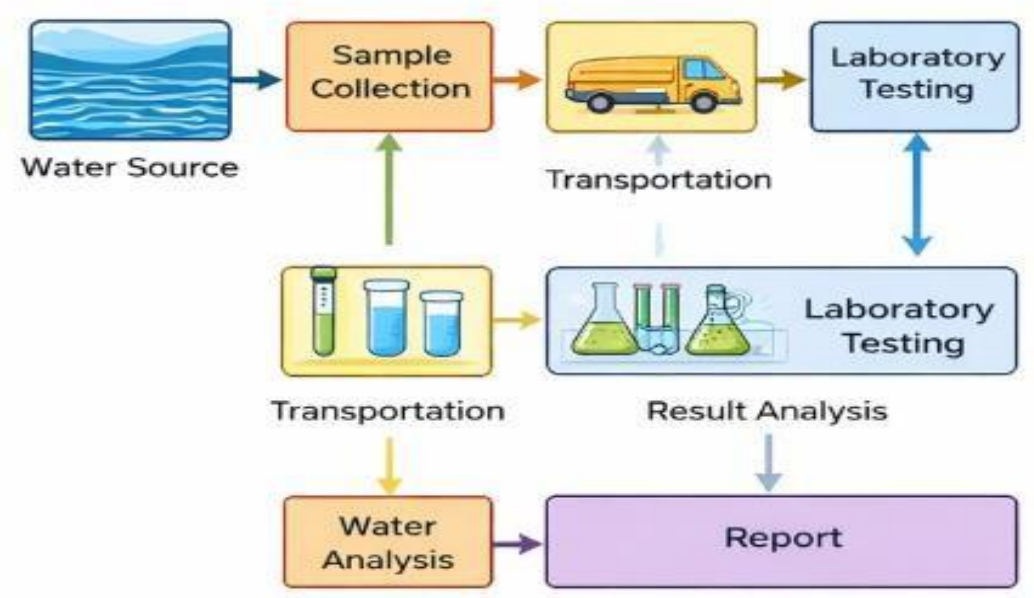


Fig 2.1.2: Water Quality Testing Process

Existing water quality monitoring systems mainly rely on traditional laboratory testing methods. These methods require collecting water samples and sending them to laboratories for detailed analysis using specialized equipment. Although these methods provide accurate results, they are time-consuming, expensive, and require skilled technicians. Because of these reasons, it is difficult to perform regular water quality testing, especially in rural and remote areas.

Some modern systems use sensors and IoT technology to monitor water quality parameters such as pH, turbidity, temperature, and Total Dissolved Solids (TDS). While these systems provide real-time monitoring, they usually depend on internet connectivity and smartphones to display the results. In many remote areas, stable internet connections are not available, which limits the effectiveness of such systems.

Another limitation is that most existing systems provide alerts through mobile notifications, SMS messages, or digital displays. These alerts may not be easily understood by all users, particularly elderly people or those who are not familiar with technology. In rural areas, many people may not be able to read digital messages or operate smartphone applications.

Additionally, several existing monitoring systems are expensive due to the use of advanced sensors, cloud services, and complex hardware components. The cost of installation and maintenance makes these systems unsuitable for small communities or villages with limited financial resources. Some systems also require continuous power supply and regular calibration of sensors, which can be difficult to manage in remote locations.

In addition, many existing systems are expensive to install and maintain because they use advanced hardware components, cloud storage, and complex communication technologies. This makes them unsuitable for small villages and low-income communities. Some systems also require continuous electrical power and regular maintenance, which can be difficult in remote areas where power supply may be unstable.

Another limitation is that most systems focus only on monitoring and data collection rather than providing simple and immediate alerts to people at the water source. Without a clear warning system, people may continue to use contaminated water unknowingly.

Furthermore, many systems focus only on data collection and monitoring but do not provide immediate and simple alerts to the users at the location. Without clear alerts, people may continue to use contaminated water without knowing its condition.

Therefore, there is a need for a low-cost, easy-to-use, and reliable water quality monitoring system that can operate in remote areas without depending heavily on internet connectivity. A system with voice announcement alerts can effectively inform users about water safety, making it more accessible and practical for rural communities.

2.2 Proposed System

Fig. 2.2 shows the Drinking Water Quality Monitoring and Alert System. The system uses different sensors integrated with an ESP32 microcontroller to continuously monitor water quality parameters such as pH, turbidity, and TDS in real time. When any parameter exceeds the predefined safe limits, the system generates alerts through a buzzer and voice announcements. Additionally, the collected data is transmitted to the Blynk IoT platform for remote monitoring and visualization.

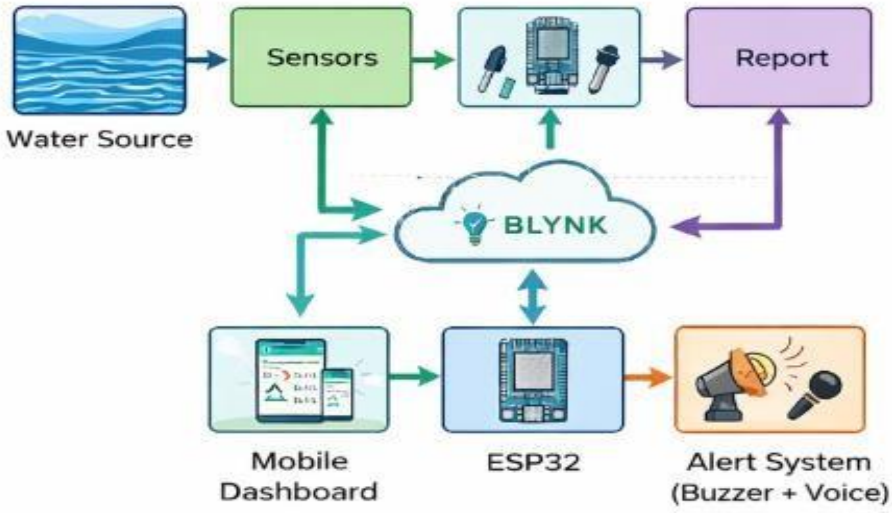


Fig.2.2: Drinking Water Quality Monitoring and Alert System

The proposed system is designed to monitor the quality of drinking water and provide immediate alerts when the water becomes unsafe. The system uses sensors to measure important water parameters such as pH level, turbidity, and temperature. These sensors are connected to a microcontroller, such as ESP32 or Arduino**,** which collects and processes the sensor data in real time.

The microcontroller continuously compares the measured values with predefined safe limits for drinking water. If the water quality is within the acceptable range, the system indicates that the water is safe for consumption. If any parameter exceeds the safe limit, the system detects that the water may be contaminated and triggers an alert.

One of the main features of the proposed system is the voice announcement module**,** which provides audio alerts through a speaker. This allows users to easily understand whether the water is safe or unsafe without needing to read text messages or use mobile applications. The system is particularly useful for people living in rural and remote areas where internet connectivity and digital literacy may

be limited.

The proposed system is designed to be low-cost, easy to install, and simple to operate, making it suitable for rural communities, villages, and small water supply systems. By providing continuous monitoring and clear alerts, the system helps ensure the safety of drinking water and improves public health in remote areas.

Furthermore, the system operates automatically without requiring manual intervention. Once the sensors are placed in the water source, the system continuously collects data and analyzes the water quality parameters. The voice alert system immediately informs users when there is any change in the water condition. This quick response helps prevent people from consuming contaminated water.

The proposed system also supports scalability, meaning that additional sensors such as TDS sensors, dissolved oxygen sensors, or conductivity sensors can be added in the future to improve water quality analysis. The system can also be powered using renewable energy sources such as solar panels, making it more suitable for remote areas where electricity supply may be limited.

Overall, the proposed system provides an efficient, reliable, and user-friendly solution for drinking water quality monitoring. By combining sensor technology, microcontroller processing, and voice announcement alerts, the system helps create a safer water supply system for rural and remote communities.

2.2.1 Advantages of Proposed System

The proposed system offers several advantages in monitoring and maintaining the quality of drinking water. One of the main advantages is real-time monitoring, which allows continuous observation of water quality parameters such as pH, turbidity, and temperature. This helps in detecting contamination immediately and prevents the consumption of unsafe water.

Another important advantage is the voice announcement feature, which provides clear audio alerts to inform users whether the water is safe or unsafe for drinking. This is especially helpful in rural and remote areas where people may not be able to read digital displays or use smartphones. The voice alert system ensures that everyone can easily understand the water quality status.

The system is designed to be low-cost and easy to implement, making it suitable for small communities and villages with limited financial resources. It uses simple electronic components and sensors, which reduces the overall cost of installation and maintenance. The system also performs automatic monitoring, which reduces the need for manual testing and human intervention.

Another advantage is that the system can be expanded and upgraded in the future. Additional sensors such as TDS sensors, dissolved oxygen sensors, or conductivity sensors can be added to improve water quality analysis. The system can also be connected to **I**oT platforms for remote monitoring and data storage, allowing authorities to track water quality from different locations.

The proposed system also helps in early detection of water contamination, which can prevent waterborne diseases and protect public health. Since the system provides quick alerts, users can take immediate action to avoid using contaminated water. Additionally, the system can be installed in water tanks, wells, borewells, and community water supply systems, making it useful for a wide range of applications.

Overall, the proposed system provides a reliable, user-friendly, and efficient solution for monitoring drinking water quality. By combining sensor technology, microcontrollers, and voice alerts, the system ensures better water safety and supports healthier living conditions in rural and remote communities.

2.3 Software & Hardware Requirements

2.3.1 Software Requirements

- **Operating System:** Windows
- **Programming Environment:** Arduino IDE
- **Microcontroller Programming:** ESP32 / Arduino Programming
- **Libraries and Frameworks:** WiFi Library, Blynk Library, DHT Sensor DFPlayer Mini Library
- **IoT Platform:** Blynk Cloud Platform
- **Database Management**: Blynk Cloud Database
- **Visualization and Analytics Tools:** Blynk Mobile Application Dashboard

2.3.2 Hardware Requirements

- **Microcontroller:** ESP32 Development Board
- **pH Sensor:** Used to measure the acidity or alkalinity of water
- **Turbidity Sensor:** Used to detect the clarity or cloudiness of water
- **Voice Module:** DF Player Mini Voice Module for audio alerts
- **Speaker:** Used to provide voice announcements
- **Power Supply:** 5V Power Adapter / Battery
- **Connecting Components:** Breadboard, Jumper Wires, Resistors
- **Water Container:** For testing and monitoring water samples

2.4 Feasibility Study

2.4.1 Technical Feasibility

The proposed Drinking Water Quality Alert System with Voice Announcement is technically feasible because it uses easily available electronic components such as ESP32/Arduino, water quality sensors (pH, turbidity, and temperature), and a voice module. These components can be easily integrated and programmed using Arduino IDE. The sensors continuously monitor water parameters, and the microcontroller processes the data to determine whether the water is safe for drinking. When unsafe conditions are detected, the system generates a voice alert through a speaker. The ESP32 also supports WiFi connectivity, which allows the system to connect with IoT platforms like Blynk for remote monitoring. Since the required hardware and software tools are simple, affordable, and widely available, the system can be easily implemented and maintained, making it suitable for real-world use in rural and remote areas.

2.4.2 Robustness & Reliability

The proposed Drinking Water Quality Alert System is designed to be robust and reliable for continuous monitoring of water quality. The system uses durable sensors and a stable microcontroller such as ESP32 or Arduino to ensure accurate measurement of water parameters like pH, turbidity, and temperature. The system continuously collects and processes sensor data, allowing it to quickly detect any changes in water quality. It is designed to operate with minimal human intervention and can function effectively in different environmental conditions. The voice announcement feature ensures that users receive clear alerts whenever unsafe water conditions are detected. Since the system uses simple and reliable hardware components, it can provide consistent performance with low maintenance, making it suitable for long-term use in rural and remote areas.

2.4.3 Economic Feasibility

The proposed Drinking Water Quality Alert System is economically feasible because it uses low-cost and easily available components such as ESP32/Arduino, pH sensor, turbidity sensor, temperature sensor, and a voice module. These components are affordable and do not require expensive installation or maintenance. The system is designed to provide continuous monitoring of water quality without the need for frequent laboratory testing, which helps reduce operational costs. Since the system is simple to build and maintain, it is suitable for rural and remote areas where financial resources may be limited. Overall, the proposed system offers a cost-effective solution for monitoring drinking water quality and ensuring public health safety.

CHAPTER-3

ARCHITECTURAL DESIGN

3.1 Modules Design

3.1.1 Water Quality Sensing Module:

This module is responsible for collecting data from the water source. It includes sensors such as the pH sensor, turbidity sensor, and temperature sensor. The pH sensor measures the acidityor alkalinity of the water, the turbidity sensor detects the clarity or presence of suspended particles, and the temperature sensor measures the water temperature. These sensors continuously monitor the water conditions and send the measured data to the microcontroller for further processing.

3.1.2 Data processing Module:

This module is the main control unit of the system and consists of a microcontroller such as ESP32 or Arduino. It receives the sensor readings and processes the data using a programmed algorithm. The microcontroller compares the sensor values with predefined safe limits for drinking water. Based on this comparison, the system decides whether the water is safe or unsafe. This module controls the overall operation of the system and coordinates the functioning of all other modules.

3.1.3 Voice Alert Module:

The voice alert module is responsible for providing audio notifications to users. It consists of a DFPlayer Mini voice module and a speaker. When the system detects that the water quality parameters exceed safe limits, the microcontroller activates the voice module. The speaker then announces a warning message such as “Water is not safe for drinking.” This feature is especially useful for users who may not be able to read digital displays

3.1.4 IoT Monitoring Module:

The IoT monitoring module allows the system to send water quality data to an IoT platform such as Blynk through WiFi connectivity. Using the Blynk mobile application, users or authorities can monitor the water parameters remotely in real time. This module helps in tracking water quality trends and maintaining proper records for future analysis.

3.1.5 Dashboard Power Supply Module:

The power supply module provides the required electrical power to operate all components in the system. It may use a 5V power adapter, battery, or rechargeable power source. A stable power supply ensures that the sensors, microcontroller, and other modules function continuously without interruption.

3.1.6 Data Display and Monitoring Module:

This module displays the real-time sensor values and system status through the IoT application or optional display units. It allows users to easily observe the water quality parameters and understand the condition of the water.

3.2 Methods & Algorithm Design

3.2.1 Data Collection

Water quality data is collected using sensors such as pH sensor, turbidity sensor, and temperature sensor placed in the water source. These sensors continuously measure the water parameters and send the readings to the microcontroller (ESP32/Arduino). The collected data helps in determining whether the water is safe or unsafe for drinking.

`

3.2.2 Data Preprocessing

The sensor data is processed to ensure accuracy and reliability. Basic preprocessing steps such as sensor calibration, noise filtering, and value stabilization are applied to obtain correct readings. This helps the system analyze the data effectively and avoid incorrect alerts.

3.2.3 Model Training and Evaluation

The microcontroller processes the sensor values and compares them with predefined safe drinking water limits. If the values remain within the safe range, the system continues monitoring. If any parameter exceeds the safe limit, the system detects contamination and triggers the alert mechanism.

3.2.4Visualization and Interpretation

The system allows monitoring of water quality parameters through an IoT platform such as Blynk. Users can view sensor readings and system status using a mobile application dashboard. This helps in understanding water quality trends and monitoring the system remotely

3.2.5 Voice Alert Mechanism:

The system includes a voice announcement module (DFPlayer Mini with speaker) to provide audio alerts. When unsafe water conditions are detected, the system generates a voice message informing users that the water is not safe for drinking. This feature ensures that people in rural areas can easily understand the warning without reading digital messages

3.3 Project Architecture

3.3.1Architectural Diagram

Fig.3.3.1 Depicts the architectural diagram of the Drinking Water Quality Alert System with Voice Announcement. It illustrates how different system components interact to monitor water quality and generate alerts. The system is organized into several layers including the sensor layer, processing layer, communication layer, alert system, and power supply.

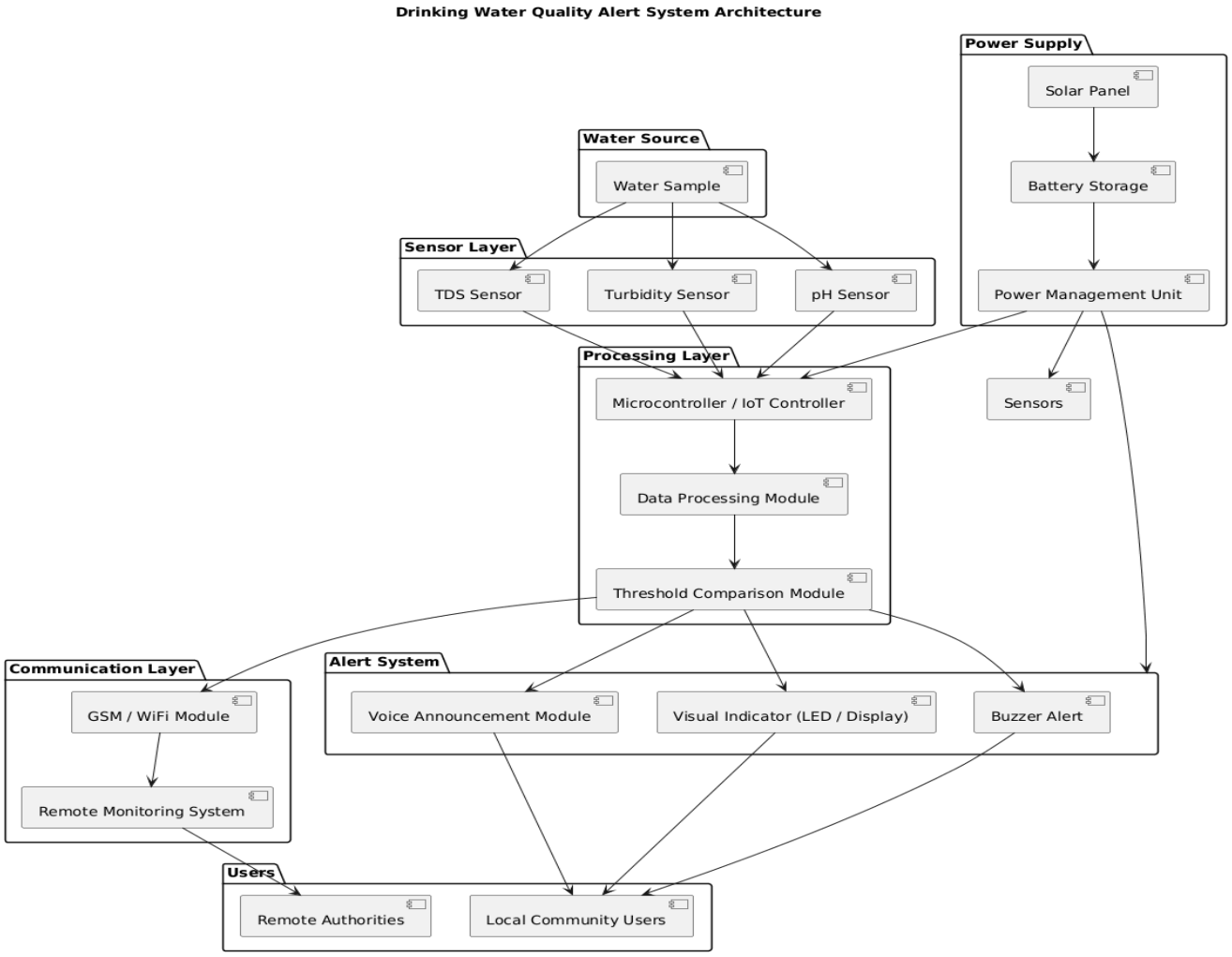


Figure 3.3.1 Architectural Diagram of Water Quality Alert System

Water Source: The system begins with the water source, where a water sample is collected for analysis. This sample may come from wells, borewells, water tanks, rivers, or community water supply systems. The water sample is then analyzed using different sensors to determine its quality.

Sensor Layer

The sensor layer consists of sensors such as TDS sensor, turbidity sensor, and pH sensor. These sensors continuously monitor important water parameters.

The TDS sensor measures the amount of dissolved solids in water.

The turbidity sensor measures the cloudiness or clarity of water.

The pH sensor measures the acidity or alkalinity of water.

The sensor readings are transmitted to the processing layer for further analysis.

Processing Layer

The processing layer contains the microcontroller or IoT controller (such as ESP32 or Arduino). This component acts as the brain of the system. It receives data from the sensors and processes the information using programmed algorithms. The data processing module analyzes the sensor readings and sends them to the threshold comparison module, where the values are compared with standard safe limits for drinking water.

Alert System

If the sensor values exceed the safe limits, the alert system is activated. The alert system consists of multiple notification mechanisms such as:

Voice announcement module – provides audio messages through a speaker to inform users about unsafe water conditions.

Visual indicator (LED or display) – shows the system status visually. Buzzer alert – generates an audible alarm for immediate warning.

Communication Layer

The communication layer uses GSM or WiFi modules to transmit the system data to a remote monitoring system. This allows authorities or users to monitor water quality remotely using mobile applications or IoT dashboards.

Power Supply

The system is powered through a solar panel and battery storage system. The power management unit regulates and distributes power to the sensors, microcontroller, and other components. This setup makes the system suitable for remote areas where continuous electricity supply may not be available.

Users

The final output of the system is directed to users, which include remote authorities and local community members. They receive alerts and monitoring data, enabling them to take necessary actions when water becomes unsafe for consumption.

Overall, the architectural design ensures efficient water quality monitoring, real-time alert generation, and remote monitoring capability, making it a reliable solution for ensuring safe drinking water in rural and remote areas.

3.3.2Data Flow Diagram

Fig. 3.3.2 shows the data flow diagram, which illustrates how water quality data moves through the system for monitoring and alert generation. Water samples are analyzed using pH, turbidity, and TDS sensors, and the data is sent to the ESP32 microcontroller for processing. The system performs threshold comparison to determine whether the water is safe or unsafe. The processed data is transmitted to the cloud platform via Wi-Fi/GSM for storage and remote monitoring. If unsafe conditions are detected, the system activates alerts such as a buzzer, LED, and voice announcement.

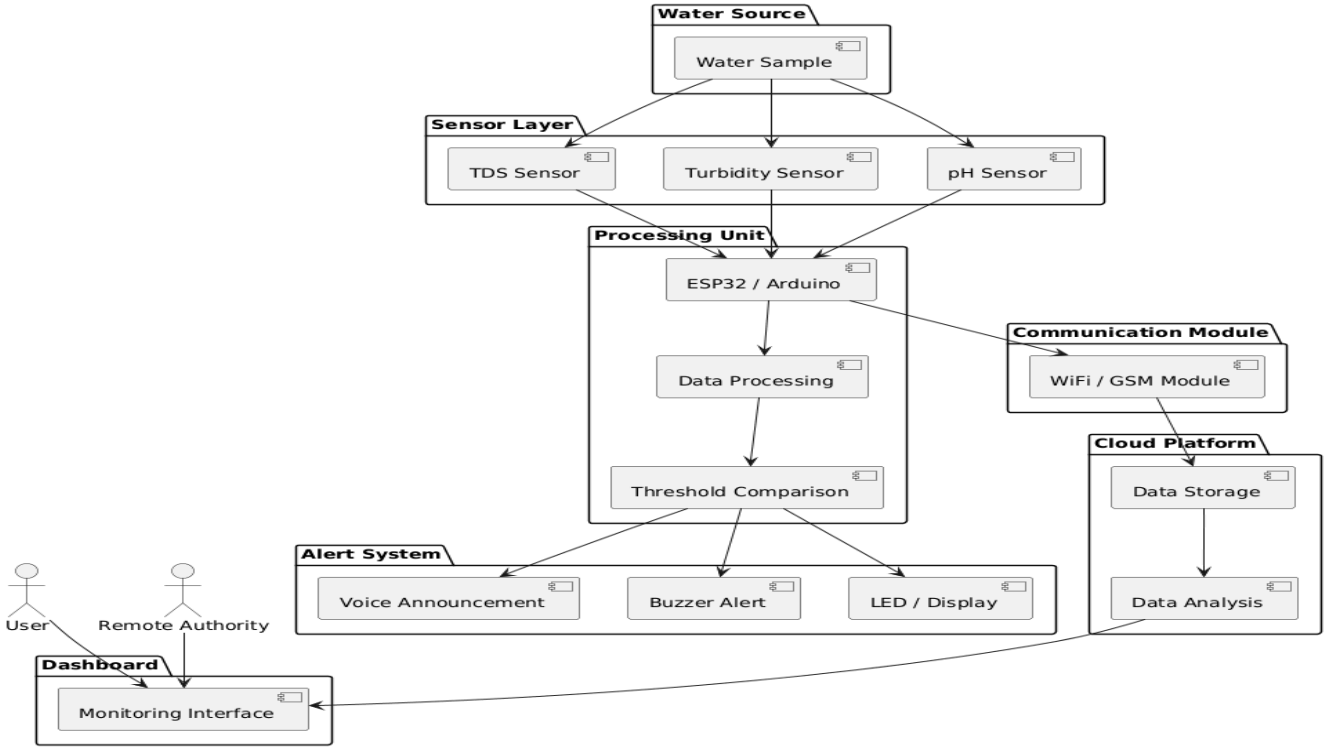


Figure 3.3.2 Data Flow Diagram of Water Quality Alert System

Water Source:

The process starts with the water source, where the water sample is taken for quality monitoring. This sample is used to analyze the condition of drinking water**.**

Sensor Layer:

The water sample is monitored by the sensor layer, which includes TDS sensor, turbidity sensor, and pH sensor. These sensors measure important parameters such as dissolved solids, water clarity, and acidity of the water.

Processing Unit:

The sensor readings are sent to the processing unit, which contains the microcontroller (ESP32/Arduino). The microcontroller collects data from all sensors and performs data processing to interpret the readings.

Threshold Comparison:

After processing the sensor data, the system performs a threshold comparison. The measured values are compared with predefined safe limits to determine whether the water is safe or unsafe for drinking.

Alert System:

If the water quality parameters exceed the safe limits, the alert system is activated. The alert system provides warnings through LED/display indicators, buzzer alerts, and voice announcements to notify users about unsafe water conditions.

Communication Module:

At the same time, the processed data is sent to the communication module, which includes a WiFi or GSM module. This module transfers the system data to the cloud platform for remote monitoring.

CloudPlatform

The cloud platform stores the transmitted data in the data storage module and performs data analysis to evaluate water quality information.

Dashboard Monitoring:

The analyzed data is displayed on the dashboard monitoring interface. Both users and remote authorities can access the dashboard to view real-time water quality data and system alerts.

This data flow ensures that the system provides continuous monitoring, real-time alerts, and remote access to water quality information, helping users quickly identify unsafe drinking water.

3.3.3 Class Diagram

Fig. 3.3.3 depicts the class diagram, which represents the structure of the water quality monitoring system and its main components. The base class Sensor is extended by pH, turbidity, and TDS sensors, which collect water quality data. These sensors send data to the microcontroller, which processes and transmits it using the Wi-Fi module. The data is stored and analyzed on the cloud platform, enabling monitoring through the dashboard. The alert system checks threshold values and triggers notifications, including voice announcements, when unsafe conditions are detected.

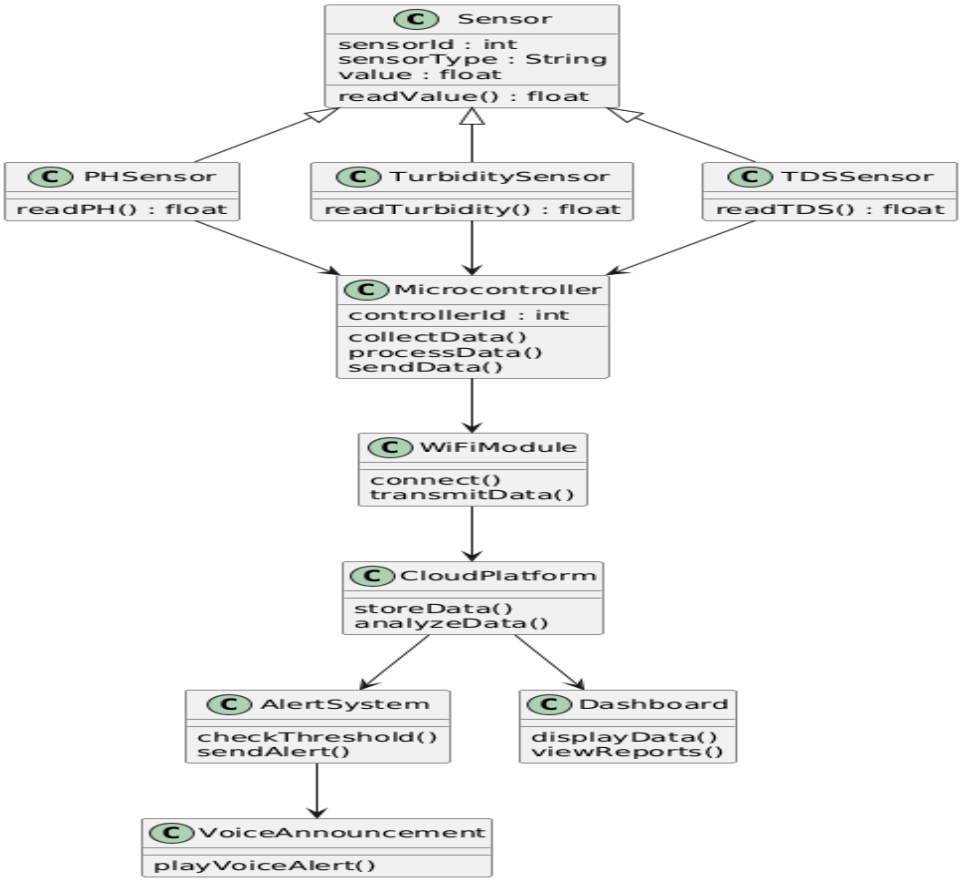


Figure 3.3.3 Class Diagram of Water Quality Monitoring System

Sensor Class:

The Sensor class is the base class of the system. It contains attributes such as sensor Id, sensor Type, and value. The function read Value () is used to read the sensor data from the water source.

pH Sensor, Turbidity Sensor, and TDS Sensor

The PHSensor, Turbidity Sensor, and TDS Sensor classes are derived from the Sensor class. Each classmeasures a specific water parameter. The readPH() function reads the pH level, readTurbidity() measuresthe clarity of water, and readTDS() measures the total dissolved solids present in water.

Microcontroller Class:

The sensor data is sent to the Microcontroller class, which acts as the main control unit of the system. It contains the attribute controllerId and functions such as collectData(), processData(), and sendData(). The microcontroller collects the sensor readings, processes the data, and prepares it for transmission**.**

WiFi Module:

The processed data is sent to the WiFi Module class. This module establishes the network connection using the connect function and transmits the collected data to the cloud platform using transmitData().

Cloud Platform:

The CloudPlatform class stores and analyzes the transmitted data. It contains functions such as storeData() for storing the data and analyzeData() for analyzing water quality information.

Alert System:

After analyzing the data, the AlertSystem class checks whether the water parameters exceed safe limits using the checkThreshold() function. If unsafe conditions are detected, the system generates alerts through the sendAlert() function.

Voice Announcement:

The Voice Announcement class provides audio alerts to users. It uses the playVoiceAlert() function to produce voice messages informing users about unsafe water conditions.

Step 8: Dashboard:

Finally, the Dashboard class displays the monitored data and reports to users. Functions such as displayData() and viewReports() allow users to observe water quality information and system status through a monitoring interface.

3.3.4 Use case Diagram

Fig. 3.3.4 shows the use case diagram, which illustrates the interaction between different actors and the water quality monitoring system. The administrator is responsible for setting threshold values, configuring sensors, and monitoring system status. The sensor system collects water parameters and processes the data before uploading it to the cloud. The system analyzes the data and generates alerts when unsafe conditions are detected. The user can view water quality data and receive alerts for timely action.

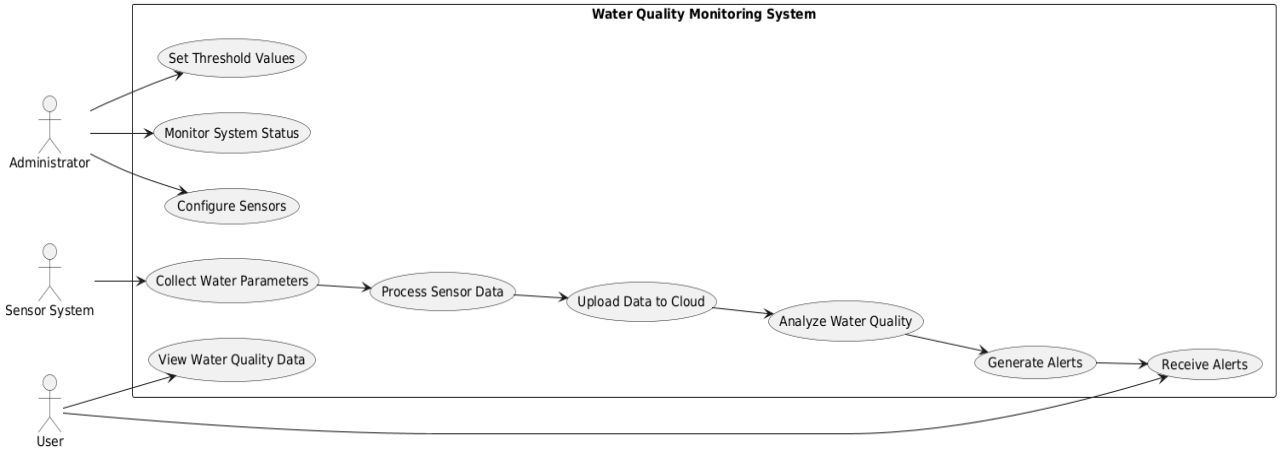


Figure 3.3.4 Use Case Diagram of Water Quality Monitoring System

Administrator Interaction:

The Administrator manages and controls the water quality monitoring system. The administrator can set threshold values for water quality parameters such as pH, turbidity, and TDS. They can also configure sensors and monitor the system status to ensure that the system is functioning properly.

Sensor System Operation:

The Sensor System is responsible for collecting water quality parameters from the water source. Sensors measure values such as pH level, turbidity, and total dissolved solids. The collected data is then sent for further processing.

Processing Sensor Data:

After collecting the sensor readings, the system processes the sensor data using the microcontroller. This step prepares the data for analysis and ensures the values are correctly interpreted.

Upload Data to Cloud:

The processed data is then uploaded to the cloud platform using a communication module such as WiFi or GSM. This allows remote monitoring and storage of water quality data.

Analyze Water Quality:

The cloud system analyzes the water quality data by comparing the measured values with predefined threshold limits. This helps determine whether the water is safe or unsafe for consumption.

Generate Alerts:

If the water quality exceeds safe limits, the system generates alerts. These alerts notify users about unsafe drinking water conditions.

User Interaction

The User can view water quality data through a monitoring interface or dashboard. When alerts are generated, the user receives alerts through voice announcements, buzzer notifications, or other alert mechanisms.

This use case diagram shows how administrators, sensors, and users interact with the system to monitor water quality, analyze data, and receive alerts when unsafe conditions are detected

3.3.5 Sequence Diagram

Fig. 3.3.5 shows the sequence diagram, which illustrates the step-by-step interaction between system components during water quality monitoring. Sensors collect data and send it to the ESP32 microcontroller for processing and threshold checking. If the water is safe, the system updates the dashboard and displays the normal status to the user. If unsafe conditions are detected, the alert system is triggered to notify the user through a buzzer or voice alerts. The processed data is also sent to the cloud for storage and remote monitoring through the dashboard.

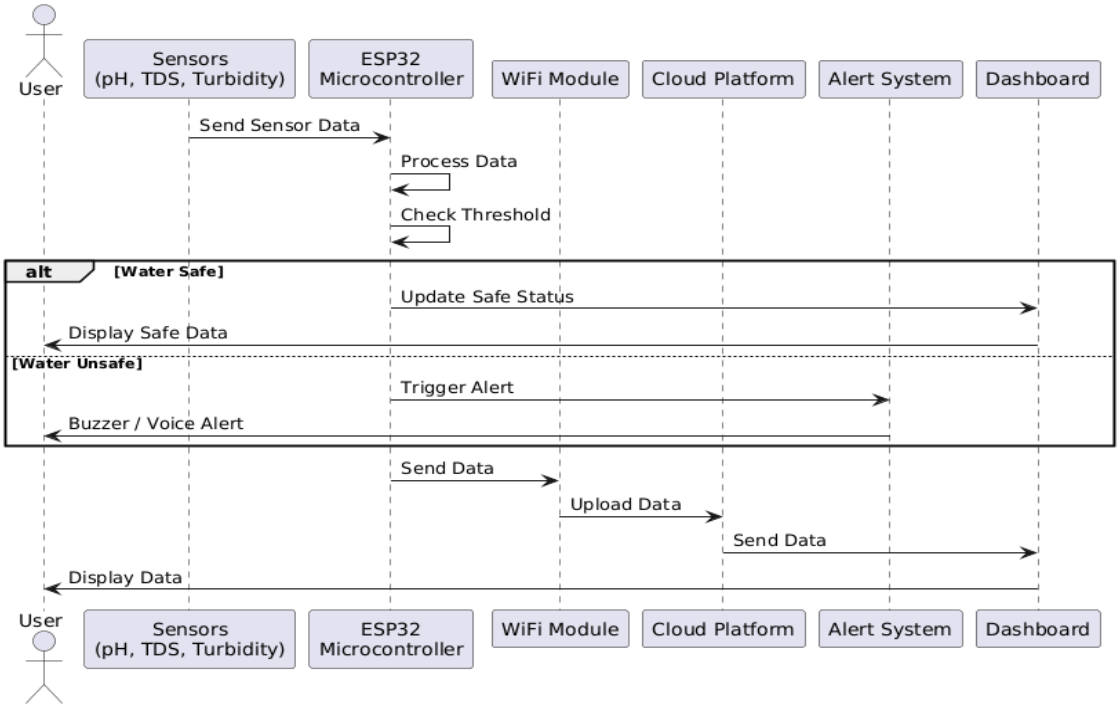


Figure 3.3.5 Sequence Diagram of Water Quality Monitoring System

Water Quality Sensors

The process begins with the water quality sensors, which measure important water parameters such as pH, TDS, turbidity, and temperature. These sensors continuously collect data from the water source.

Microcontroller Processing:

The collected sensor data is sent to the microcontroller (ESP8266/NodeMCU or ESP32/Arduino).
The microcontroller processes the sensor readings and prepares the data for transmission.

Data Transmission through Wi-Fi Module

After processing the data, the microcontroller sends the information to the Wi-Fi module. The Wi-Fi module establishes a network connection and transmits the data to the cloud platform.

Cloud Platform Storage and Analysis:

The cloud platform receives the uploaded data and performs data storage and analysis. The system checks whether the measured parameters exceed predefined threshold values.

Condition Check:

At this stage, the system evaluates the water quality parameters. Two possible conditions may occur: If parameters exceed the threshold values, the system identifies unsafe water conditions.

If parameters are within the normal range, the water is considered safe.

Alert SystemActivation:

If unsafe conditions are detected, the alert system is triggered. The system sends notifications or alerts to inform users about the unsafe water quality.

User Notification:

Finally, the user receives notifications or alerts about the water quality. If the parameters are normal, the system simply displays the data on the dashboard for monitoring purposes.

This sequence diagram illustrates the step-by-step interaction between sensors, microcontroller, communication module, cloud platform, alert system, and users to ensure real-time water quality monitoring and alert generation.

3.3.6 Activity Diagram

Fig. 3.3.6 depicts the activity diagram, which represents the workflow of the water quality monitoring system. The system starts by initializing the sensors and Wi-Fi connection. It continuously collects and processes water quality data, then sends it to the cloud for analysis. The system checks whether the parameters are within safe limits. If the conditions are safe, the data is displayed on the dashboard; otherwise, alerts are generated and users are notified.

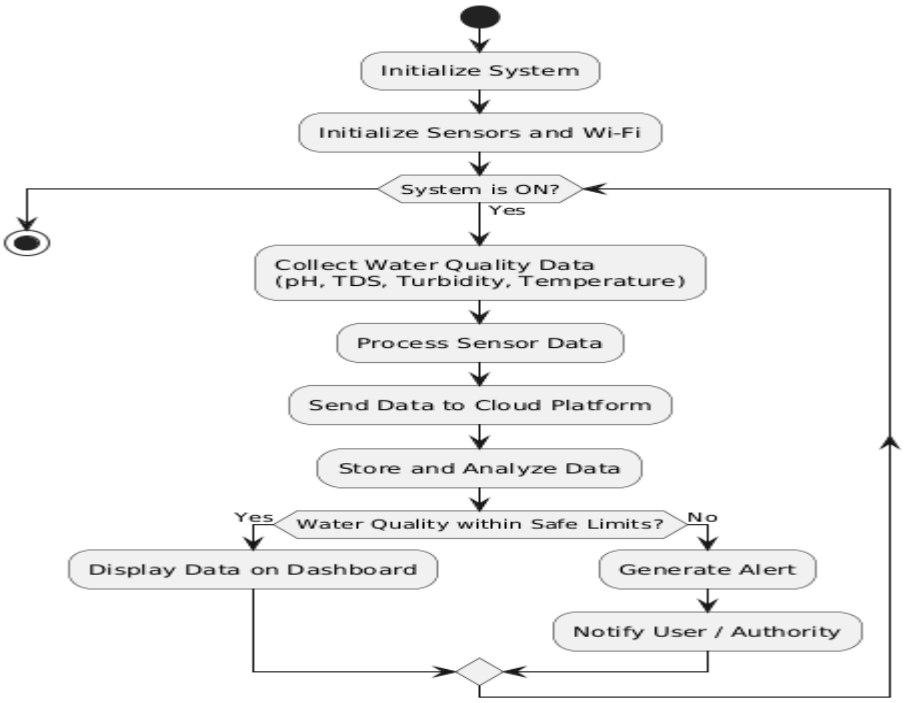


Figure 3.3.6 Activity Diagram of Water Quality Monitoring System

Initialize System:

The process starts by initializing the system, which prepares the microcontroller and other system components to begin operation.

Initialize Sensors and Wi-Fi

Next, the system initializes the water quality sensors and Wi-Fi module. This step ensures that sensors such as pH, TDS, turbidity, and temperature sensors are ready to collect data and the Wi-Fi module is ready to transmit data.

Collect Water Quality Data:

The system then collects water quality data from the sensors. The sensors measure important parameters such as pH level, TDS, turbidity, and temperature from the water source

Process Sensor Data:

After collecting the sensor readings, the microcontroller processes the data to interpret the values and prepare them for further analysis.

Send Data to Cloud Platform:

The processed data is then sent to the cloud platform through the Wi-Fi module for remote storage and monitoring.

Store and Analyze Data:

The cloud platform stores the collected data and analyzes it to determine whether the water quality parameters are within safe drinking limits.

Water Quality Condition:

At this stage, the system checks whether the water quality is within safe limits.

If the values are within the safe range, the system continues monitoring normally. If the values exceed the safe limits, the system identifies unsafe water conditions.

Display or Generate Alert:

If the water quality is safe, the system displays the data on the dashboard for users to monitor. If the water quality is unsafe, the system generates an alert.

Notify User or Authority:

When an alert is generated, the system notifies the user or responsible authority through alerts such as voice announcements, buzzer alerts, or notifications.

System Continues Monitoring:

Finally, the system remains active and continues monitoring water quality, repeating the process to ensure continuous observation and timely alerts.

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CHAPTER-4

IMPLEMENTATION & TESTING

4.1 Coding Blocks

4.1.1 Import Required Libraries

```
#define BLYNK_PRINT Serial
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <DFRobotDFPlayerMini.h>
#include <HardwareSerial.h>
```

4.1.2 Define Blynk Credentials

```
#define BLYNK_TEMPLATE_ID "TMPL3oBWx9P0u"
#define BLYNK_TEMPLATE_NAME "Water monitoring"
#define BLYNK_AUTH_TOKEN "YOUR_AUTH_TOKEN"
```

4.1.3 Configure WiFi Setting

```
char ssid[] = "Hii";
char pass[] = "12345678";
```

4.1.4 Define Sensor and DFPlayer Pins

```
#define PH_PIN 34
#define TURB_PIN 35
#define TDS_PIN 32
#define DF_RX 16
#define DF_TX 17
```

4.1.5 Define Safe Limits

```
float ph_min = 6.5;
float ph_max = 8.5;
float turb_max = 5.0;
float tds_max = 500.0;
```

4.1.6 Calibration Parameters

```
float ph_offset = 0.0;
float turb_clear_voltage = 2.75;
float tds_factor = 0.13;
```

4.1.7 Initialize Modules

```
HardwareSerial dfSerial(2);
DFRobotDFPlayerMini player;

BlynkTimer timer;
```

4.1.8 Voltage Averaging Function

```
float readVoltage(int pin)
{
  long sum = 0;
  for(int i=0; i<30; i++)
  {
    sum += analogRead(pin);
    delay(5);
  }
  float avg = sum / 30.0;
  return avg * (3.3 / 4095.0);
}
```

4.1.9 Sensor Data Processing

```
void sendData()
{
  float phVoltage = readVoltage(PH_PIN);
  float turbVoltage = readVoltage(TURB_PIN);
  float tdsVoltage = readVoltage(TDS_PIN);

  float pH = 7 + ((2.5 - phVoltage) * 3.5) + ph_offset;
  if(pH < 0) pH = 0;
  if(pH > 14) pH = 14;

  float turbidity = (turb_clear_voltage - turbVoltage) * 300;
  if(turbidity < 0) turbidity = 0;

  float tds_raw = (tdsVoltage / 2.3) * 1000;
  float tds = tds_raw * tds_factor;
```

4.1.10 Display Serial Output

```
  Serial.println("------ Water Data ------");
  Serial.print("pH: "); Serial.println(pH);
  Serial.print("Turbidity: "); Serial.println(turbidity);
  Serial.print("TDS: "); Serial.println(tds);
  Serial.println("------------------------");
```

4.1.11 Send Data to Blynk Cloud

```
Blynk.virtualWrite(V0, pH);
Blynk.virtualWrite(V1, turbidity);
Blynk.virtualWrite(V2, tds);
```

4.1.12 Water Quality Evaluation and Voice Alert

```
int errorCode = 0;

if(pH < ph_min || pH > ph_max) {
  errorCode = 4;
}
else if(turbidity > turb_max) {
  errorCode = 5;
}
else if(tds > tds_max) {
  errorCode = 3;
}

if (errorCode != 0)
{
  Serial.println("Status: UNSAFE");
  Blynk.virtualWrite(V3, 1);

  if (errorCode != lastErrorCode || (millis() - lastAudioTime > 10000))
  {
    if (errorCode != lastErrorCode) {
      player.play(1);
      delay(1200);

      player.play(2);
      delay(1500);
    }

    player.play(errorCode);
    lastAudioTime = millis();
  }
}
else
{
  Serial.println("Status: SAFE");
  Blynk.virtualWrite(V3, 0);
}

lastErrorCode = errorCode;
}
Send Data to BlynkCloud
Blynk.virtualWrite(V0, pH);
Blynk.virtualWrite(V1, turbidity);
Blynk.virtualWrite(V2, tds);
```

4.1.13 Setup Function

```
void setup()
{
  Serial.begin(115200);

  analogReadResolution(12);
  analogSetAttenuation(ADC_11db);

  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

  dfSerial.begin(9600, SERIAL_8N1, DF_RX, DF_TX);

  if(!player.begin(dfSerial, true, false)) {
    Serial.println("DFPlayer not detected. Check connections.");
  }

  player.volume(25);

  timer.setInterval(5000L, sendData);
}

analogReadResolution(12);
analogSetAttenuation(ADC_11db);

Serial.println("System Starting...");

Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

timer.setInterval(3000L, processWaterData);
```

4.1.14 Main Loop

```
void loop() { Blynk.run();
  timer.run();

  }
```

4.2 Execution Flow

4.2.1System Start:

When the ESP32 is powered on, the program begins execution. The system initializes serial communication using Serial.begin(115200) to display monitoring data on the serial monitor.Load the Dataset:

4.2.2 Initialize Hardware Components:

The buzzer pin is configured as an output using pin Mode (BUZZER_PIN, OUTPUT), and it is initially turned off. The ESP32 analog input settings are configured using analog Read Resolution (12) and analog Set Attenuation (ADC_11db) to improve sensor reading accuracy.

4.2.3 Connect to Wi-Fi and Blynk Cloud:

The ESP32 connects to the Wi-Fi network using the provided SSID and password. After establishing the Wi-Fi connection, the device connects to the Blynk IoT platform using the authentication token. This enables remote monitoring through the Blynk mobile application.

4.2.4 Start Timer for Periodic Monitoring:

A timer is configured using timer set Interval (3000L, process Water Data), which triggers the process Water Data () function every 3 seconds to read and process sensor data.

4.2.5 Sensor Data Acquisition:

Inside the process Water Data () function, the system reads analog values from the pH sensor, TDS sensor, and turbidity sensor. The function get Avg Voltage () collects multiple samples and calculates the average voltage to reduce noise and improve measurement stability.

4.2.6 Calculate Water Quality Parameters:

The program converts sensor voltages into meaningful water quality parameters: The pH value is calculated using calibration constants.

The TDS value is calculated using a polynomial formula with temperature compensation. The turbidity value is calculated based on the difference between the measured voltage and the clear-water baseline.

4.2.7 Check Water Safety Conditions:

The system compares the calculated values with predefined safe limits. If the pH is outside the range 6.5–8.5, turbidity is greater than 5 NTU, or TDS is greater than 500 ppm, the water is classified as UNSAFE.

4.2.8 Generate Alerts:

If unsafe conditions are detected, the buzzer is activated to alert users. The system also updates the water quality status as SAFE or UNSAFE.

4.2.9 Display Data on Serial Monitor:

The program prints the measured values of pH, TDS, turbidity, and system status on the serial monitor for debugging and monitoring.

4.2.10 Send Data to Blynk Dashboard:

The calculated values are transmitted to the Blynk cloud server using virtual pins. The mobile application displays these parameters on the dashboard for real-time monitoring.

4.2.11 Continuous Monitoring Loop:

The loop () function continuously runs Blynk.run () and timer.run().

This keeps the device connected to Blynk and ensures that sensor readings and alerts are updated every few seconds for continuous water quality monitoring.

CHAPTER-5

RESULTS & DESCRIPTION

5.1 Resulting Screen

The developed IoT-based Drinking Water Quality Monitoring and Alert System with Voice Announcement was successfully implemented and tested using an ESP32 microcontroller integrated with pH, turbidity, and TDS sensors. The system continuously monitors water quality parameters in real time and transmits the collected data to the cloud platform for remote visualization through the Blynk application. The results show that the system accurately detects variations in water quality and effectively determines whether the water is safe or unsafe based on predefined threshold values. When the parameters exceed safe limits, the system immediately generates alerts through a buzzer and voice announcement, ensuring quick user awareness. The prototype demonstrates reliable performance, ease of use, and cost-effectiveness, making it suitable for deployment in rural and remote areas where continuous water quality monitoring is essential.

Fig. 5.1 depicts the prototype of the IoT-based water quality monitoring system. The entire water monitoring system is built using an ESP32, which takes pH, turbidity, and TDS sensors as its inputs. These sensors are used to measure water quality parameters in real time. The collected data is transmitted to the ESP32 microcontroller and later to a cloud server through communication modules and power supply units, as presented. This prototype provides a practical and effective insight into designing such water quality monitoring systems.



Fig.5.1: IoT-based Prototype for Monitoring Water Quality

Fig.5.2 depicts the IoT-based water quality monitoring dashboard implemented using the Blynk application. The dashboard displays real-time values of key water quality parameters such as pH, turbidity, and total dissolved solids (TDS). In the shown output, the pH value is approximately 6.93, which lies within the acceptable range for safe drinking water. The turbidity value is 0, indicating clear water with no suspended particles. The TDS value is around 83.97 ppm, which is well below the permissible limit, confirming good water quality. The LED indicator shows a safe status, providing a visual representation of water safety. This dashboard enables users to monitor water quality remotely and take timely action when necessary.

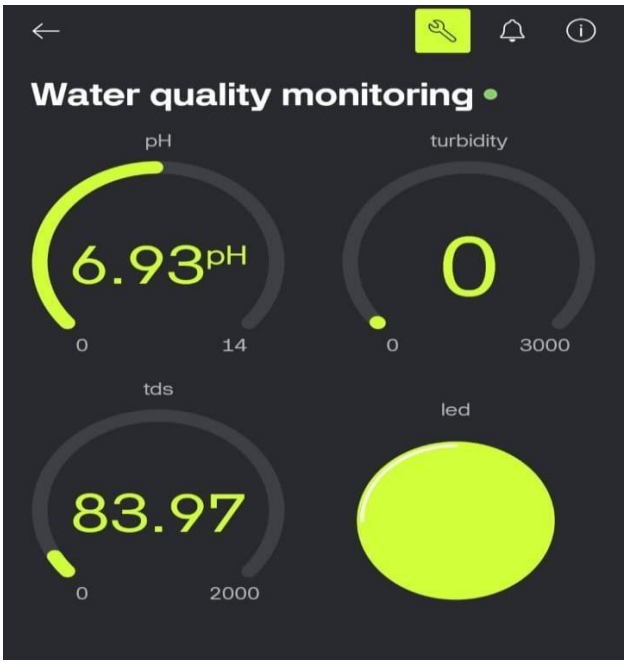


Fig.5.2 Real Time Water Quality Monitoring Dashboard

Fig.5.3 shows the serial monitor output when the system detects unsafe water quality conditions. The ESP32 microcontroller reads data from the pH, TDS, and turbidity sensors and displays the measured values on the serial monitor. In this output, the pH value is around 7.70, which is within the normal range for drinking water. However, the TDS value is approximately 584 ppm, which exceeds the recommended safe limit of 500 ppm. Additionally, the turbidity value is very high (around 108–356 NTU), indicating that the water contains a large amount of suspended particles and impurities.

Since some of the measured parameters exceed the predefined safe limits, the system classifies the water as **"UNSAFE."**

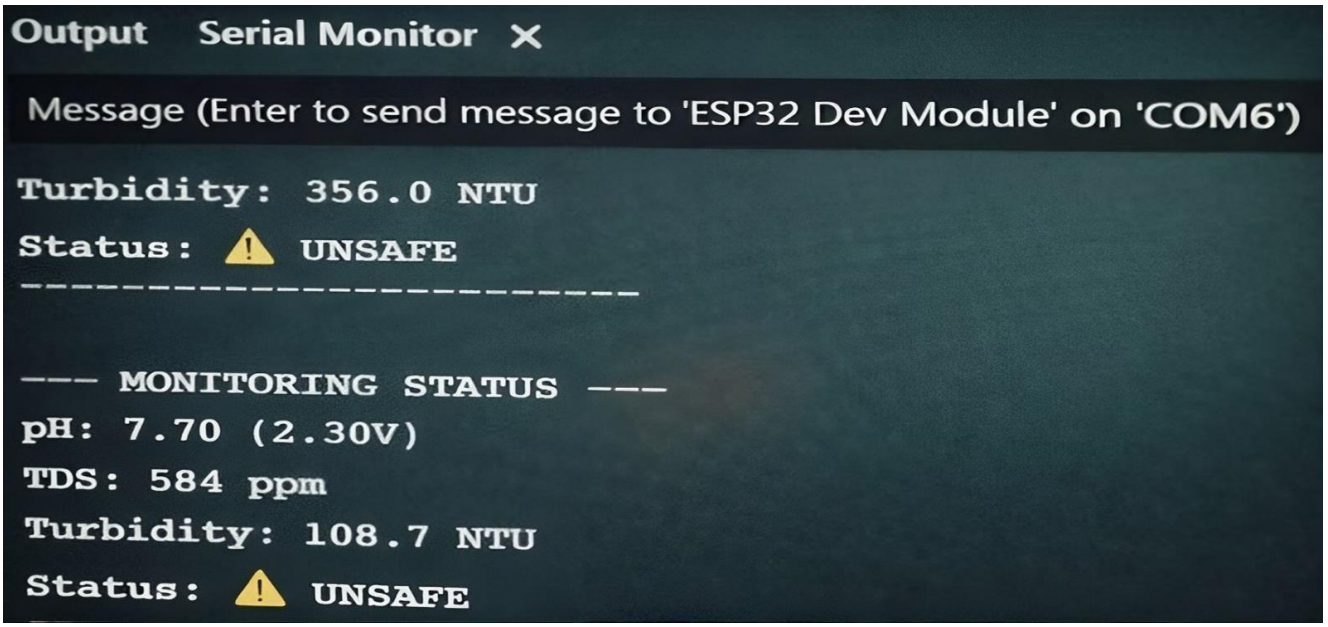


Fig.5.3 Water Quality Alert Message On PC

Fig.5.4 shows the serial monitor output of the water quality monitoring system when the water parameters are within safe limits. The ESP32 microcontroller reads sensor data from the pH sensor, TDS sensor, and turbidity sensor and displays the measured values in real time. In this output, the pH value is approximately 7.17, which indicates neutral water and falls within the safe drinking range of 6.5 to 8.5. The TDS value is around 448–450 ppm, which is also within the acceptable limit for drinking water. Additionally, the turbidity value is 0.0 NTU, indicating that the water is clear and free from suspended particles.

Since all the measured parameters are within the predefined safe limits, the system displays the status as "SAFE.

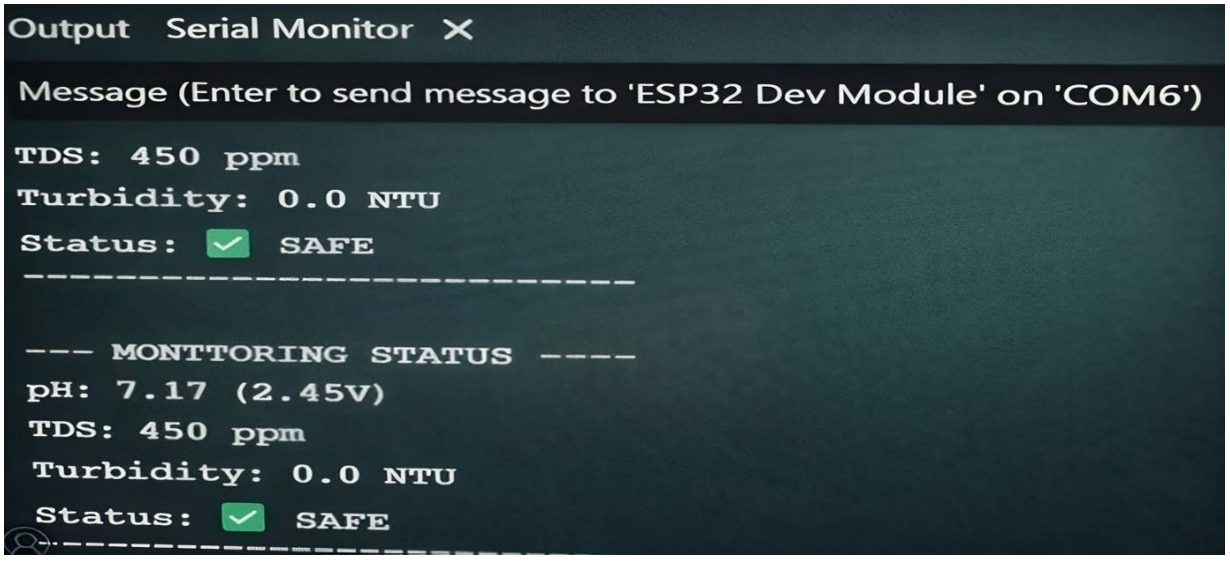


Fig.5.4Testing Results on PC When Water is Safe

CHAPTER-6

CONCLUSIONS & FUTURE SCOPE

6.1 Conclusion

In this project, an IoT-Based Drinking Water Quality Monitoring and Alert System with Voice Announcement has been successfully designed and implemented. The system uses sensors such as pH sensor, TDS sensor, and turbidity sensor to continuously monitor important water quality parameters. The collected data is processed by the ESP32 microcontroller, which evaluates whether the water quality falls within safe drinking limits.

The system also integrates IoT technology using the Blynk platform, allowing real-time monitoring of water quality through a mobile application or dashboard. Whenever the water quality parameters exceed the predefined safe limits, the system generates alerts using a buzzer and status notification, ensuring that users are immediately informed about unsafe water conditions.

The implementation results show that the system can accurately detect both safe and unsafe water conditions and display the results on the serial monitor and IoT dashboard. This system provides a low-cost, efficient, and reliable solution for monitoring drinking water quality, especially in rural and remote areas where regular water testing facilities may not be available. Overall, the developed system helps ensure safe drinking water and supports better health and environmental safety.

6.2 Future Scope

The proposed IoT-Based Drinking Water Quality Monitoring and Alert System can be further enhanced by integrating additional sensors to measure more water quality parameters such as dissolved oxygen, conductivity, and temperature for more accurate water analysis. In the future, the system can be improved by adding automatic water purification or filtration mechanisms that activate when contaminated water is detected.

The system can also be integrated with advanced cloud platforms and mobile applications to provide better data visualization, historical data analysis, and predictive monitoring using artificial intelligence or machine learning techniques. Additionally, the use of solar-powered systems can make the device more suitable for rural and remote areas where electricity supply is limited. Voice-based alerts can also be improved to support multiple languages, making the system more user-friendly for different communities. Overall, these improvements can make the system more intelligent, efficient, and widely applicable for ensuring safe drinking water.

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Design and Development of Drinking Water Alert System with Voice Announcements for Remote Areas

MALLA REDDY UNIVERSITY

Introduction

Access to safe drinking water remains a serious challenge in many remote and rural areas, where the lack of proper monitoring systems allows water contamination to go undetected, leading to the spread of waterborne diseases such as cholera and typhoid. This project presents a Water Quality Alert System designed to continuously monitor key water parameters like pH, turbidity, and temperature using sensors. By integrating IoT technology, the system analyzes data in real time to detect unsafe water conditions and sends instant alerts to users or authorities for timely action. The aim is to provide a low-cost, efficient, and reliable solution to improve water safety and protect public health in underserved regions.

Aim

The aim of this project is to design and develop a Water Quality Alert System for remote and rural areas to ensure safe and clean drinking water. The system focuses on continuously monitoring important water parameters such as pH, turbidity, and temperature using sensor-based technology. The system aims to provide real-time data analysis and quickly detect any contamination or unsafe conditions in water sources. When abnormal values are identified, the system sends instant alerts to users or concerned authorities, enabling timely action to prevent health risks. The project also aims to be low-cost, easy to deploy, and reliable, making it suitable for underserved regions with limited infrastructure. Overall, the goal is to improve water quality monitoring and promote better public health and sustainable development.

Methodology

The proposed Water Quality Alert System is designed using an ESP32 microcontroller integrated with16.1 pH, turbidity, and TDS sensors to continuously monitor water quality in real time. These sensors are immersed in the water source to measure essential parameters such as pH (indicating acidity or alkalinity), turbidity (indicating water clarity), and total dissolved solids (TDS), which reflect the concentration of dissolved impurities. The sensors collect analog signals, which are then processed by the ESP32 microcontroller and converted into digital values for analysis. The ESP32, equipped with built-in Wi-Fi, transmits this data to a cloud platform or server, where it is stored and can be accessed through a mobile or web-based interface for real-time monitoring. If any parameter exceeds the acceptable limits, the system automatically detects the anomaly and triggers an alert. The system is designed to be low-cost, energy-efficient, and easy to install, making it suitable for deployment in remote and rural areas where regular water quality monitoring is difficult.

Under the Guidence of :

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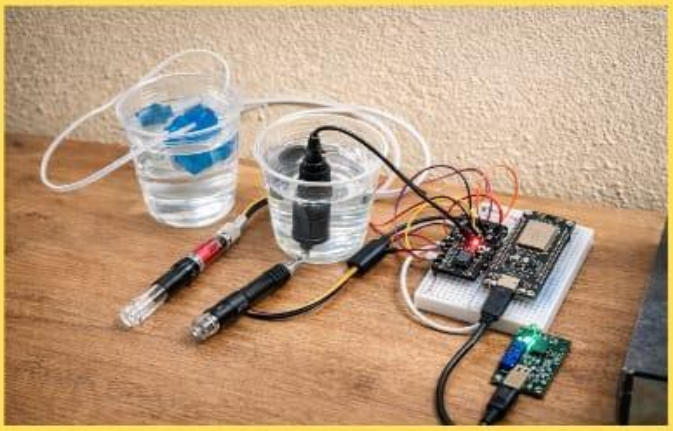
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Result

The Water Quality Alert System effectively monitors water quality in real time using pH, turbidity, and TDS sensors with the ESP32. The data is transmitted to the Blynk platform and displayed on a mobile interface The system accurately detects unsafe conditions when values exceed predefined limits and sends instant alerts to users. The results show that the system is reliable, responsive, and suitable for continuous monitoring. It provides a simple and efficient solution for ensuring safe water quality, especially in remote areas.The system allows users to track historical data trends, helping in better analysis of water quality over time. Its real-time monitoring and alert features ensure timely intervention, making it a practical solution for maintaining safe water and improving overall community health awareness in remote regions.





Conclusion

The Water Quality Alert System provides an effective and reliable solution for monitoring water quality in real time using ESP32 and sensor technology. It helps in early detection of contamination and sends instant alerts through the Blynk platform, enabling quick action to prevent health risks. The system is low-cost, easy to implement, and suitable for remote areas with limited infrastructure. Overall, it improves water safety, supports public health, and contributes to sustainable water management.

RESEARCH PAPER

Design and Development of Drinking water quality Alert system with voice announcement for Remote Areas

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Abstract- **Due to rising levels of water contamination and pollution, ensuring safe drinking water is an essential public health necessity. Traditional techniques for testing the quality of water are labor-intensive, manual, and do not yield data in real time. The development of a drinking water quality monitoring and alert system that allows for ongoing, real-time water quality parameter monitoring is presented in this project. Essential parameters like pH, turbidity, total dissolved solids (TDS), and drinking water temperature are measured by the system using a variety of sensors. These sensors are connected to a Wi-Fi-capable microcontroller, which gathers and sends sensor data to a cloud platform for processing and archiving. When anomalous conditions are found, users or relevant authorities receive immediate alerts that compare the monitored values with accepted safe drinking water limits. This system helps prevent water- borne illnesses, minimizes human labor, and guarantees prompt detection of water contamination. The suggested solution is affordable, scalable, and appropriate for applications involving rural water supplies and smart cities. By making sure that everyone, even those with low literacy levels, receives important warnings, the voice alert feature increases community awareness and response times. This device can be employed in rural and remote areas for long period since the system is made to work 24/7 with little or no human intervention. IoT makes the data to be visually represent to view live data through a dashboard or mobile applications so that authorities and residents can have water quality data with them no matter what location they are in.**

Keywords- **- IoT for Water Quality Monitoring using ESP32 with pH, TDS, and Turbidity sensors, coupled with an Alert System that is voice controlled and also pushes notificataions on Blynk.**

I.INTRODUCTION

In the home, agricultural, and industrial sectors, water contamination and inadequate quality monitoring continue to be significant problems that frequently result in health hazards and environmental deterioration. Manual sampling and laboratory analysis are the mainstays of traditional water testing techniques, which are laborious and do not allow for real-time monitoring [1]. Modern Internet of Things (IoT) technology places a strong emphasis on ongoing remote monitoring through the use of cloud platforms and smart sensors, greatly increasing productivity and reaction time [2]. Research has indicated that the integration of wireless communication modules with microcontrollers facilitates automated data collection and real-time transmission to cloud dashboards for analysis and visualization [3][4]. Numerous studies have examined Internet of Things (IoT)- based water quality monitoring systems, emphasizing automation's ability to increase accuracy, scalability, and minimize human intervention [5, 6, 7]. The proposed system fills a clear research gap because, despite these advancements, many existing systems mainly concentrate on parameter monitoring and cloud visualization, with little attention paid to integrated multi-parameter analysis, real- time SMS alerts, and low-cost scalable deployment for rural areas. Many existing systems do not provide proper user-friendly alert features to the non-technical rural community users. The most common solutions are built on mobile applications or internet connection which can be an unreliable factor in most of the remote areas. Thus to remove such drawbacks of existing solutions a Low-Cost real-time monitoring with a voice announcement feature and decision- making based alert mechanism system to check multiple parameters has been proposed.

II. LITERATURE SURVEY

IoT-based automation of environmental monitoring systems has been the subject of much recent research, especially in the area of water quality evaluation. A fundamental strategy for the real-time measurement of water parameters like pH, turbidity, and Total Dissolved Solids (TDS) is the combination of microcontrollers and smart sensors, which greatly lessens reliance on manual sampling techniques [1]. According to studies, wirelessly enabled controllers, such as the ESP32, facilitate smooth data transfer to cloud platforms, allowing for ongoing remote monitoring in both residential and commercial settings [2]. A number of studies concentrate on intelligent cloud-based visualization and alert systems that go beyond simple monitoring. Web dashboards and IoT platforms have been used to provide automated alerts when safety thresholds are exceeded as well as graphical representation of water parameters [3][4]. When compared to conventional laboratory-based testing systems, these methods show increased efficiency, quicker reaction times, and improved reliability. Additional research suggests that automated alert systems enhance public health safety by offering immediate notifications during contamination incidents [5][6].

In water monitoring systems, sensor calibration and accuracy enhancement have also become important research topics. Numerous studies look at filtering algorithms and signal conditioning strategies to improve measurement accuracy in a range of environmental circumstances [7][8]. Low-cost embedded solutions that can achieve acceptable accuracy levels for real-world deployment—even in rural and resource-constrained areas—are presented in existing literature. It has been demonstrated that incorporating several sensors into a single monitoring framework increases the system's overall robustness and capacity for multi-parameter analysis [9].Despite these developments, existing systems mainly concentrate on isolated parameter monitoring or simple cloud visualization, providing little assistance for SMS-based alert systems, scalable low-cost deployment models, and integrated multi-parameter analytics. Furthermore, many studies pay relatively little attention to quantitative evaluation metrics like performance validation, threshold optimization, and long-term data logging [10]. The foundation of the proposed research is the need for an integrated smart water quality monitoring system that integrates real-time sensing, cloud integration, automated alert mechanisms, and workable deployment strategies.

III. SYSTEM ANALYSIS

A. EXISTING SYSTEM

Fig.1: Shows Our current approach consists of manual water sampling and analysis in the laboratory. Water samples from various sources are manually drawn and laboratory analyzed to determine the quality parameters (e.g., pH, TDS, turbidity). The process is not only laborious but it does not involve real-time analysis, there is no alert system.

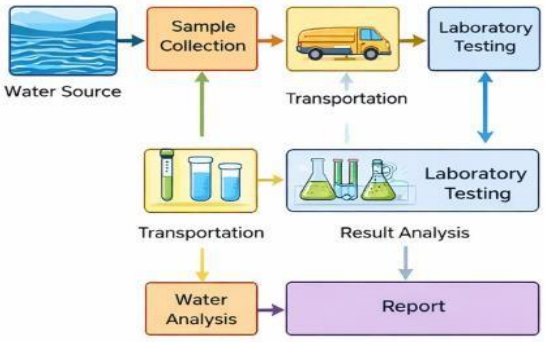


Fig.1: Water Quality Testing Process

The present water quality monitoring process mainly rely on manual sampling method and laboratory testing method. Water samples are collected from river, lake, well, borehole and tank in accordance to the procedure and then are brought to laboratory to test water parameters like pH, turbidity, temperature, and total dissolved solid (TDS). This system can bring accurate result. However, manual sampling is time consuming and laborsome work including: taking the sample, transporting, analyzing the water sample and then making the report. The time gap from the time of sample taking till getting the results makes water quality information non- reliable as the water quality itself can vary over this time and the taken water sample is not suitable for real-time analysis.

Another problem associated with this approach is lack of real-time process. This traditional manual system only collected samples at a given time, therefore, it can not detect unexpected contamination or unexpected water pollution. It demands the involvement of skilled people, the collected samples can only be carefully processed and handled manually to minimize potential human error or risk, and the operational cost is higher. Moreover, the system has no intelligent automation and analysis on its own, for example, no automated system compares water results with safety limits, and no trends are analysed based on accumulated history data.

B. PROPOSED SYSTEM

Fig.2: Shows we are using different sensors with ESP32, which can sense the water parameters like pH, turbidity and TDS values constantly to check quality of water in real-time. When any value crosses safe range we create alerts and alerts on buzzer, voice and we send this data to the Blynk IoT.

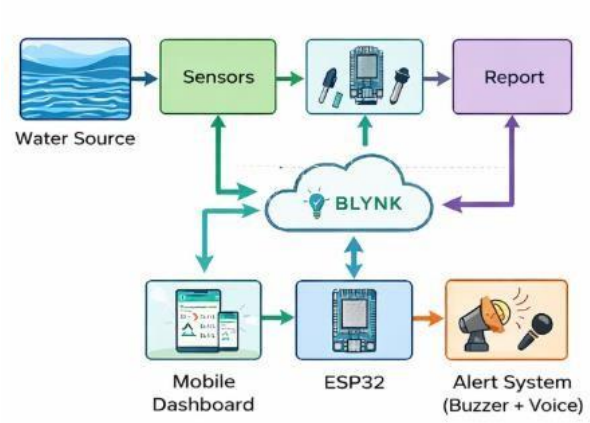


Fig.2:DrinkingWaterQualityMonitioring And Alert System

The proposed system enhances public safety and response time by coupling an Internet of Things (IoT)-based smart water quality monitoring system with a real-time voice alert mechanism. The system uses an ESP32 microcontroller to manage embedded sensors that continuously collect water quality data (such as pH, turbidity, temperature, and total dissolved solids - TDS) for the source and points of delivery. As opposed to conventional measurement methods, the monitored data is pushed into a cloud platform for long-distance data retrieval and management, analysis, storage, and usage. In case any parameter exceeds defined safety margins, the system will instantly trigger a buzzer, broadcast a voice announcement for local warning, and simultaneously notify the users via an Internet-connected mobile application; thereby the alerts can also be perceived in areas lacking internet service. Additionally, it permits for sustained storage of the data from all the sensor readings to perform historically data analysis.

Moreover, through the integration of elementary data analytical methods and an anomaly detection approach, uncommon water quality variations and sudden change can also be detected by identifying patterns, anomalies and unexpected patterns and alerting to any imminent threat. It has a flexible and extendable system architecture so that further sensors and sophisticated techniques (such asmachine learning for prediction of potential contamination events in the future) may be further appended in the coming days to achieve better results in the process. As the cost of system is minimal, energy effective and user friendly it is also ideally fit for remote and rural area; so, the system can implement affordable hardware elements (e.g. ESP32 micro-controller) to minimize device costs, thereby the most suitable options in such remote locations for providing warnings at reasonable cost and for ease to operation and maintenance even among low literacy level peoples by virtue of its voiced alerts .

IV. METHODOLOGY

Module 1: Sensor Data Collection Module

It employs the use of various water quality sensors located at the water source to determine the essential parameters of drinking water, which include pH, Turbidity, and Total Dissolved Solids (TDS).

◦ The system initializes all sensors during the startup process and continuously reads the analog signals produced by the sensors.

◦ The raw analog data is then converted to measurable units (pH value, NTU, ppm) using calibration formulas.

◦ The processed data from the sensors is transmitted to the data processing and cloud communication modules.

Module 2: Data Processing & Calibration Module

The raw sensor output may include fluctuations and noise based on environmental factors.

◦ This module involves analog-to-digital conversion and the use of calibration equations to transform voltage measurements into normalized water quality data.

◦ The processed data is compared to pre-defined safe limits based on WHO guidelines for drinking water quality.

◦ The validated data is transmitted for cloud transmission and alert evaluation.

Module 3: IoT Communication Module

◦ To enable remote monitoring of water quality in real time.

◦ The Node MCU connects to Wi-Fi and establishes communication with the Blynk IoT cloud platform.

◦ Sensor values are transmitted using virtual pins to the mobile dashboard.

◦ The system ensures continuous data updates at fixed intervals (e.g., every 3 seconds).

◦ In case of unsafe water detection, an event notification is triggered on the Blynk mobile application.

Module 4 : Threshold Evaluation & Decision Module

This module checks whether the water is safe or contaminated.

◦ The processed sensor values are compared with the threshold values:

pH level: 6.5 – 8.5 TDS: ≤ 500 ppm Turbidity: ≤ 5 NTU

◦ If any of the values go beyond the safe limits, the water is declared “Unsafe.”

◦ The result is transmitted to the alert module for further processing.

◦

Module 5: Voice Alert & Buzzer Notification Module

◦ To issue immediate local warnings at the water source point.

◦ Upon detection of unsafe water, the DFPlayer Mini module is responsible for playing a voice message through a speaker.

◦ A buzzer is also activated at the same time.

Module 6 Monitoring & User Interface Module

◦ Offers real-time graphical representation of water quality values on the Blynk mobile app.

◦ Offers real-time graphical and numeric representation of pH, turbidity, and TDS values.

◦ Allows for remote monitoring from anywhere with internet access.

Allows for storage of data logs in the cloud for future analysis

V. RESULTS

Fig.1: Shows the complete architecture of an IoT-enabled water monitoring system
The entire water monitoring system is built using an ESP32 which takes pH, turbidity and TDS sensors as its inputs. These sensors are used to measure the amount of water in real-time. The collected data is transmitted to the ESP32 micro-controller and later to cloud server through transmission and power supplies as presented. This prototype provides a real and useful insight into designing such systems.

Each sensor helps find the water quality in this application setup pH sensors indicate whether water is acid or alkaline; turbidity sensors assess water clearness and TDS sensor measures dissolved solids to make sure the quality is suitable for drinking. These sensors have been fully submersed in water samples.

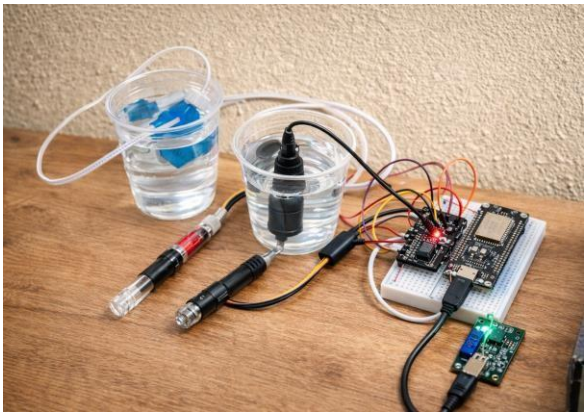


Fig.1: IoT-based Prototype for Monitoring Water Quality

Fig. 2 depicts the IoT-based water quality monitoring dashboard implemented in the Blynk mobile application. On the dashboard, we can visualize the key parameters of water like pH, turbidity, and total

dissolved solids (TDS) in real-time.

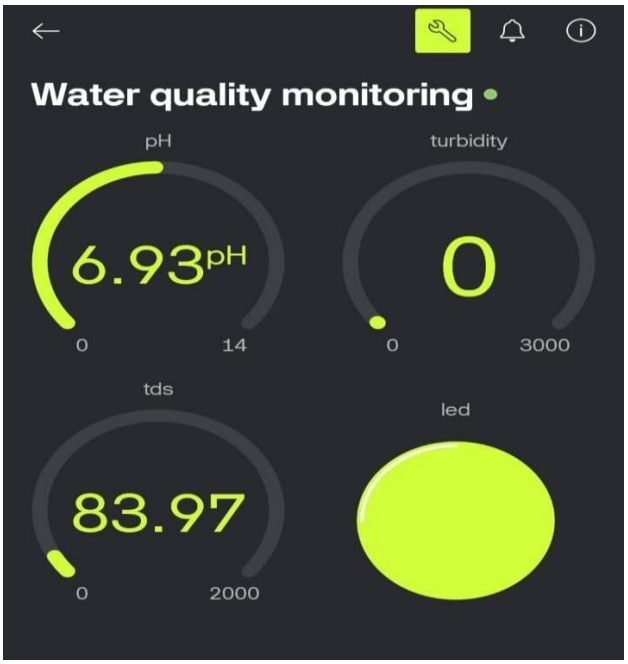


Fig.2: Blynk Dashboard Showing Parameters of Water Quality

Fig. 3 shows the serial monitor output for safe limits of water quality parameters. The water quality system real time shows the values of pH, TDS and Turbidity from the sensors and based on the parameters they fall in safe range the output is SAFE which means the water is safe to drink and use.

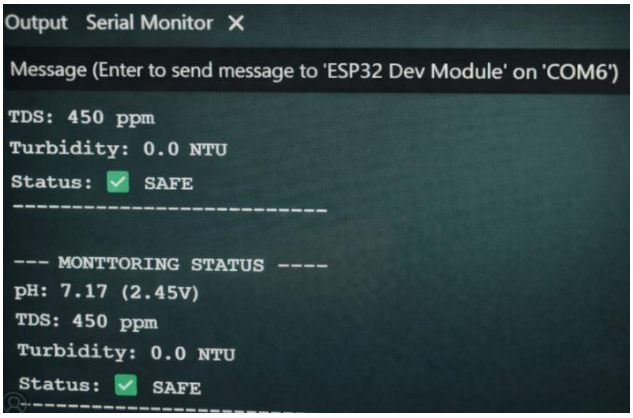


Fig.3: Safe water condition

Fig.4 Shows about the serial monitor output when the water quality exceeds safe limits. The system displays real-time values of pH, TDS, and turbidity, where the turbidity level is high, indicating contamination. As a result, the system marks the water status as UNSAFE. This output confirms that the system can effectively detect abnormal conditions and alert users immediately. The serial monitor helps in monitoring sensor readings and verifying the system's response during unsafe situations

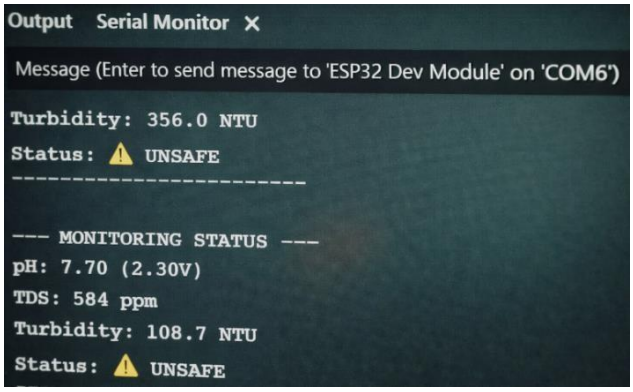


Fig.4: Unsafe water condition

The **Fig.5** present the serial monitor output displaying continuous water quality readings over time. The system records parameters such as pH, turbidity, and TDS at regular intervals. The values indicate abnormal conditions, where pH is low and turbidity and TDS levels are high, suggesting poor water quality.
This output demonstrates the system's ability to perform continuous monitoring and log real-time data, helping in identifying changes in water quality and detecting contamination effectively

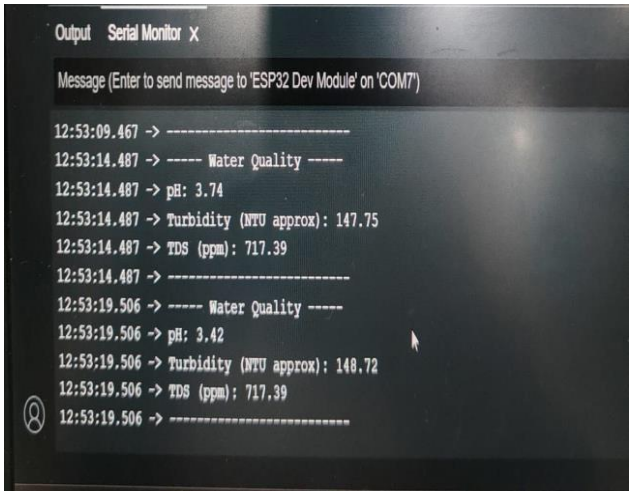


Fig.5: Water Quality Parameters Serial Monitor

VI. CONCLUSION:

In this project, an IoT-Based Drinking Water Quality Monitoring and Alert System with Voice Announcement has been successfully designed and implemented. The system uses sensors such as pH sensor, TDS sensor, and turbidity sensor to continuously monitor important water quality parameters. The collected data is processed by the ESP32 microcontroller, which evaluates whether the water quality falls within safe drinking limits. The system also integrates IoT technology using the Blynk platform, allowing real-time monitoring of water quality through a mobile application or dashboard. Whenever the water quality parameters exceed the predefined safe limits, the system generates alerts using a buzzer and status notification, ensuring that users are immediately informed about unsafe water conditions.

The implementation results show that the system can accurately detect both safe and unsafe water conditions and display the results on the serial monitor and IoT dashboard. This system provides a low-cost, efficient, and reliable solution for monitoring drinking water quality, especially in rural and remote areas where regular water testing facilities may not be available.

Overall, the developed system helps ensure safe drinking water and supports better health and environmental safety

Moreover, the system is highly dependable and capable to regularly survey parameters like water quality along with negligible human interruption. It enables fast tracing out of any anomaly using the obtained and processed data as and when there is occurrence of anything unusual.

VII. Future Scope:

Additional sensors can be integrated into the proposed IoT-based water quality monitoring system to detect water quality more Parameters including Dissolved Oxygen (DO), Temperature, Biological Contaminants can be additionally incorporated to provide the comprehensive water quality status with great efficiency and accuracy. Features like ML to forecast water quality; predict water contamination; providing solar power as supply for the system .SMS,mobile appbased water quality reports, government monitoring; smart city infrastructure based on water quality information can also be introduced in the future to improve the overall system and provide more benefits. Furthermore, by using web based advanced monitoring dashboard, the decision can be taken based on data analysis and historical information and also by improving the voice indication system in multilingual format helps a diverse range of public users. By implementing those, the system would more effective, reliable, scalable, intellectual, and user friendly.

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Water contamination is soaring high. It is extremely important for us to find something to test our drinking water and determine if it is safe or not. Manual testing requires labor, it is slow and does not display the output instantly. This project involves setting up an IoT enabled drinking water quality monitoring and alert system that continuously checks the parameters of the water on a timely basis. The proposed system works on pH, TDS, turbidity and temperature measurements with a multitude of sensors. These sensors are integrated and interfaced to a Wi-Fi enabled micro-controller where it checks the quality of water parameters to match the safety standards and logs them onto cloud for later reference. An instant alert will be pushed when the condition deviates from the pre-set thresholds and any abnormal activity is detected. This system aims to reduce water borne illnesses and human labor by identifying water contamination at real-time. This system is scalable and economically affordable for both rural areas and smart cities. An additional benefit of the device is a voice alert feature that will immediately inform users about issues so they can prevent harm, thus assisting in overcoming the communication barrier for users with low literacy rates and ensuring rapid public response times. This gadget can be utilized in villages as long as for the current application and the system will consistently check parameters like the quality of the water in any case of emergency so human intervention is rarely necessitated and this system saves them from drinking tainted water. IoT allows visual presentation of the collected information so authorities as well as locals have accessibility to water quality data anywhere by browsing the dashboard or utilizing mobile apps. The collected water quality data is stored history for potential use with subsequent scrutiny.

Keywords: IoT for Water Quality Monitoring using ESP32 with pH, TDS, and Turbidity sensors, coupled with an Alert System that is voice controlled and also pushes notifications on Blynk.

Contamination and subpar quality in water still remains to be a serious problem in the house, farm, and industry, bringing about well-being risks and environmental degradation, and standard techniques of testing quality have dependably utilized manual inspection and lab examination that is slow, difficult, and does not report information in real-time . In IoT, all IoT activities involve pervasive local area observation utilizing shrewd sensors and cloud stages to enhance reaction rates and productivity Numerous specialists have considered the communication part of IoT and the application in which the data acquired is given for use at a cloud framework to create a web dashboard. Research has explored and confirmed that the framework comprises IoT based water quality following systems which can help in automating workWith IoT becoming quite modern, it has helped with its ability to allow better monitoring and has had significant progress and application in numerous different fields like agricultural, domestic and medical etc. But most of the systems that were previously established on the data monitoring part and cloud visualization, do not consider to monitor multiple parameters on the same dashboard, provide alerts to all such parameters with real-time sms and have no scope to deploy this system to the rural areas at low cost. Many people from rural background is not well literate so it could be possible they miss the water quality updates of their respective tanks if they rely on notifications over the mobile app as it may or may not get data on time.